

A Russian Perspective on South Korea's Risky Nuclear Ambitions

By Vladislav Chernavskikh & Anton Khlopkov

South Korea would have much to lose by pursuing a nuclear weapons program, a hypothetical idea that its president raised recently. It would entail Seoul withdrawing from the Nuclear Non-Proliferation Treaty, triggering a cascade of negative consequences. For Russia, that would effectively halt bilateral co-operation on peaceful uses of nuclear energy.

It's time for international stakeholders to renew efforts to reduce tensions on the Korean Peninsula and create a sustainable regional security architecture with the long-term goal to free the entire peninsula from nuclear weapons, write Vladislav Chernavskikh and Anton Khlopkov.

IN JANUARY this year, South Korean President Yoon Suk-yeol said that due to the “growing threat” from North Korea, Seoul might consider deploying US tactical nuclear weapons in the country or pursuing its own military nuclear program.¹ The latter openly contradicts South Korea's obligations under the Treaty on the Non-Proliferation of Nuclear Weapons (NPT).² Yoon later dialed back the statement and assured the media that for now South Korea intends to adhere to its NPT obligations, but his initial statement has already provoked lively discussions on the risks of further nuclear proliferation on the Korean Peninsula.

WHERE ARE WE NOW?

At present, there is no indication that Seoul is pursuing a military nuclear program or taking practical steps toward that goal. But there seems to be a critical mass of external and internal factors that are compelling nonproliferation experts to pay closer attention to South Korea's nuclear policy.

First, it is reasonable to assume that Seoul might have incentives to consider further enhancing its military capabilities, including a possible move toward possession of a “super-weapon.” The security environment on the Korean Peninsula is tense, and a worrying trend toward further escalation has taken form in recent months. North Korea continues to develop its nuclear and missile programs — in late December 2022, the leader of North Korea (the Democratic People's Republic of Korea), Kim Jong Un, announced plans to begin mass production of tactical nuclear weapons and to further build up the country's nuclear arsenal.³

¹ “Yoon keeps possibility of Seoul's possession of nuclear weapons open,” *Korea Times*, Jan 12, 2023. www.koreatimes.co.kr/www/nation/2023/01/356_343477.html

² “Yoon reaffirms commitment to nuclear treaty in Davos,” *Yonhap News*, Jan 20, 2023. en.yonhap.com/view/AEN20230120004100315

³ “Report on 6th Enlarged Plenary Meeting of 8th WPK Central Committee,” *Korean Central News Agency*, Jan. 1, 2023, www.kcna.kp/en/article/q/e6667bf3afa7c65e0939f77ee83eabf9.kcmsf

⁴ “Austin in South Korea as Yoon seeks greater US nuclear assurances,” *Reuters*, Jan. 31, 2023, www.asahi.com/ajw/articles/14828098

⁵ “Over 76% of South Koreans support development of nuclear weapons,” *Korea Times*, Jan 30, 2023. www.koreatimes.co.kr/www/nation/2023/01/205_344519.html

⁶ “DPRK Strategic Capabilities and Security on the Korean Peninsula: Looking Ahead,” a joint study by the Center for Energy and Security Studies and the International Institute for Strategic Studies, July 14, 2021, www.iiss.org/blogs/research-paper/2021/07/dprk-strategic-capabilities-security-korean-peninsula

⁷ Peter Hayes, “Park Chung Hee, the CIA & the Bomb,” *Global Asia*, September 2011 (Vol. 6, No. 3), www.globalasia.org/v6no3/feature/park-chung-hee-the-cia-&-the-bomb_peter-hayes

Attempts to reinvigorate regional security dialogue involving the key stakeholders — the former participants in the Six-Party Talks — have been unsuccessful so far. Against this backdrop, elements of South Korea's political establishment and some experts are increasingly vocal in asking how much reliance can be placed on the US “nuclear umbrella” in the event of an open military conflict in the region.⁴

Second, there is no doubt that South Korea has sufficient scientific, technical, industrial and economic capacity. In speaking about the possibility of the country acquiring its own nuclear arsenal, President Yoon emphasized that if the political decision were made, nuclear weapons could be produced in a relatively short amount of time due to already existing expertise. According to some sources, representatives from South Korea's nuclear industry speak about being able to produce nuclear weapons within six months of a political decision to do so.

Third, support for the acquisition of nuclear weapons continues to increase steadily among South Koreans. According to the latest South Korean public opinion polls, three out of four people are in favor of the country possessing its own nuclear weapons.⁵

Fourth, South Korea's nonproliferation record is far from perfect historically. In 1973, South Korea developed a long-term plan to produce nuclear weapons. Incidentally, information about the nuclear activities in South Korea under President Park Chung Hee was one of the factors that led to Pyongyang's sustained interest in pursuing its own nuclear program.⁶ South Korea's clandestine military nuclear program went on for several years even after its accession to the NPT and was finally rolled back in the late 1970s under US pressure.⁷ Still, South Korea continued working with sensitive nuclear technology. According to the International Atomic Energy Agency (IAEA),

between 1982 and 2000 the country carried out activities related to uranium conversion, chemical and laser enrichment (including production of highly-enriched uranium) and plutonium separation, all of which were not reported to the IAEA, contrary to South Korea's obligations under the Comprehensive Safeguards Agreement.⁸ It's worth remembering that in 2002-03 Seoul denied IAEA inspectors the right to access the experimental uranium enrichment facility in South Korea as well as to collect environmental samples.⁹ Furthermore, in 1997-2001, South Korean agents were known to have attempted to illegally obtain components for intercontinental ballistic missiles capable of carrying nuclear warheads.¹⁰

HYPOTHETICAL SCENARIOS AND OUTCOMES

It's reasonable to assume that as long as the South Korea-US alliance is seen by Seoul as a critical element of national security, the US has the most effective political leverage to influence the nuclear policy of the South Korean leadership. However, Washington has not always been able to prevent its allies from developing nuclear weapons — the example of Pakistan comes to mind.

South Korea could make a political decision to develop and produce nuclear weapons, either through a clandestine military nuclear program or an open withdrawal from the NPT. A combination of these is also hypothetically possible.

In the first case, sooner or later, the IAEA is likely to find indications that Seoul is not complying with its safeguards agreement. Such indications may include anomalies in the accounting data of the reported nuclear material or detection of undeclared nuclear material, facilities and activities. This would be followed by an IAEA investigation and a report to the IAEA Board of Governors. If South Korea refuses to comply with the IAEA requirements regarding noncompliance, Russia in co-operation with other countries

If Seoul withdrew from the NPT, it could also deal a serious blow to South Korea's energy sector and economy. As a result of likely sanctions, its nuclear power plants, which provided 27 percent of the country's electrical power generation in 2022 and are fully dependent on imported uranium and uranium enrichment services, would be at risk of shutting down.

⁸ "Implementation of the NPT Safeguards Agreement in the Republic of Korea," Report by the Director General, IAEA, Nov 11, 2004, www.iaea.org/sites/default/files/gov2004-84.pdf

⁹ "South Korea's Nuclear Surprise," *Bulletin of the Atomic Scientists*, Vol. 61, No. 11 (2005), pp. 40-49, doi.org/10.2968/061001011

¹⁰ "South Korea Missile Overview," Nuclear Threat Initiative, Apr 21, 2016, www.nti.org/analysis/articles/south-korea-missile/; "A businessman's confession: I brought ICBMs into Korea," *Chosun Ilbo*, Aug. 1, 2020 (in Korean), www.chosun.com/site/data/html_dir/2011/06/24/2011062401222.html

¹¹ UN Security Council, S/RES/1737 (2006), www.un.org/securitycouncil/s/res/1737-%282006%29

¹² Treaty on the Non-Proliferation of Nuclear Weapons, UN Office for Disarmament Affairs, www.un.org/disarmament/wmd/nuclear/npt/text/

¹³ George Bunn and Roland T Imerbaev, "The Right to Withdraw from the Nuclear Non-Proliferation Treaty: the views of two NPT negotiators," originally published in Russian in *Yaderny Kontrol*, Vol. 111, No.3 (Fall 2005), www.files.ethz.ch/isn/123371/2005_Right_to_Withdraw_from_NPT.pdf

¹⁴ World Nuclear Performance Report 2022, World Nuclear Association, www.world-nuclear.org/getmedia/9dafaf70-20c2-4c3f-ab80-f5024883d9da/World-Nuclear-Performance-Report-2022.pdf.aspx

will likely support the subsequent transfer of the South Korean nuclear dossier to the UN Security Council to adopt a resolution urging the country to return to compliance with its NPT obligations. In 2006, after the IAEA's board of governors adopted a resolution on Iran's noncompliance with its safeguards agreement, the UN Security Council adopted Resolution 1737 (2006),¹¹ which, among other things, decided that all states should prevent the supply and transfer of certain sensitive nuclear technology and materials to Iran.

It is worth emphasizing that, as of now, according to the data available, the IAEA has found no evidence that South Korea is not complying with its safeguards agreement. Moreover, this year, according to some sources, the IAEA strengthened its inspection activities in the country due to the overall growth of South Korean scientific, technological and industrial nuclear capabilities.

A second hypothetical option could be a South Korean announcement of its withdrawal from the NPT. According to Article X of the NPT,¹² all parties to the treaty and the UN Security Council should be notified of such a decision three months in advance. This is likely to be followed by a meeting of the three depositary states of the treaty to co-ordinate their positions. Subsequently, a South Korean withdrawal statement would be discussed at the UN Security Council, which is capable of taking action against South Korea if its withdrawal is deemed a "threat to peace and security."¹³ Moscow would likely co-ordinate with Beijing and propose measures against Seoul to be included in the resolution, encouraging South Korea to remain in the NPT. At the first stage, these could include banning the export of materials and technologies that could contribute to Seoul's missile and nuclear programs.

One of the global consequences of any decision by South Korea to withdraw from the NPT would be a further weakening of the global nuclear non-

proliferation regime. The emergence of a new crisis would put additional pressure on the regime, reduce its chances of functioning effectively to resolve existing problems, and create risks of further proliferation of nuclear weapons. Seoul's decision to withdraw from the NPT would also probably put an end to the long history of multilateral diplomatic efforts to create conditions for the denuclearization of the Korean Peninsula.

Some analysts have also emphasized in informal conversations that the US may decide to veto a UN Security Council resolution imposing sanctions on South Korea. This would result in further deepening disagreements among the permanent UN Security Council members on nuclear nonproliferation issues and would effectively paralyze the council's ability to fulfill its responsibilities in this area.

If Seoul withdrew from the NPT, it could also deal a serious blow to South Korea's energy sector and economy. As a result of likely sanctions, its nuclear power plants, which provided 27 percent of the country's electrical power generation in 2022 and are fully dependent on imported natural uranium and uranium enrichment services, would be at risk of shutting down.¹⁴ As the country's nuclear fuel reserves become depleted, operators of South Korean nuclear power plants would likely be forced to first reduce capacity and then to shut down operations completely within three years at the latest.

Such a scenario would effectively nullify South Korea's nuclear export potential — developing nuclear exports has been named as one of the goals of the Yoon administration.¹⁵ A decision to withdraw from the NPT would also impact the US capacity to build new nuclear power plants, including small modular reactors (SMRs), since the US industry is currently significantly dependent on South Korea in manufacturing and supplying major equipment for reactors. As a result,

15 "Yoon pledges to export 10 nuclear power plants by 2030 if elected," Yonhap News Agency, Dec 29, 2021, enyna.co.kr/view/AEN20211229005400315

16 "The Statute of the IAEA," International Atomic Energy Agency, www.iaea.org/sites/default/files/statute.pdf

17 "Agreement Between the Government of the Republic of Korea

and the Government of the Russian Federation on Cooperation on the Peaceful Uses of Atomic Energy," May 28, 1999, fissilematerials.org/library/rok99.pdf

18 "Decision No. 973 of Dec. 15, 2000, on the export and import of nuclear materials, equipment, special non-nuclear materials and related technologies," Government of the Russian Federation, docs.cntd.ru/document/901777341

Russia and China would have additional opportunities in the global market.

South Korea would also lose its diplomatic influence and scientific presence in relevant international forums. It is a member of the IAEA Board of Governors, and in 2021-22 the South Korean representative chaired the board. In 2012, Seoul hosted the second Nuclear Security Summit, and in 2016-17 it chaired the Nuclear Suppliers Group. The country is actively involved in technical projects within the OECD NEA, Generation IV International Forum and other platforms. These formats of scientific co-operation would be denied to South Korea if it were to withdraw from the NPT.

It can be assumed that South Korea's intention to withdraw from the NPT, and its subsequent withdrawal, would be addressed in other relevant international forums besides the UN Security Council. Certainly, we can expect this issue to be reflected in the NPT review process. At the next Preparatory Committee session or the Review Conference itself, many states would probably insist that the draft final document include provisions condemning South Korea's actions and urging it to return to the NPT and comply with the relevant Security Council decisions.

At the IAEA, in accordance with its statute, members may advocate for a decision by the board of governors to suspend South Korea's participation in the agency's programs and projects.¹⁶ In addition, one could expect that South Korea's intention to withdraw from the NPT would be condemned in such major international gatherings as the G20, the Shanghai Co-operation Organization and BRICS.

Any technical and commercial nuclear co-operation between Russia and South Korea could be expected to be halted even if a UN Security Council resolution is blocked by the US or other permanent members of the council. According to the 1999 Intergovernmental Agreement between

Russia and South Korea on Co-operation in the Field of Peaceful Uses of Nuclear Energy, nuclear exports as well as nuclear and non-nuclear materials, installations and equipment received by Seoul under the agreement cannot be used for military purposes and must be subject to IAEA safeguards throughout their use within South Korean territory or under its jurisdiction.¹⁷ Russian export control legislation also establishes that exports of nuclear goods and technologies to non-nuclear-weapon states may be carried out on condition that the recipient country has a valid safeguards agreement with the IAEA that covers all of its nuclear activities.¹⁸

Consequently, the Russian nuclear industry would have to abandon co-operation with South Korean companies. In particular, co-operation on the Russian-constructed El Dabaa nuclear power plant in Egypt would be terminated — currently, Korea Hydro & Nuclear Power is scheduled to supply US\$2 billion worth of equipment to the Egyptian site. Shipments of enriched uranium products from Russia to South Korea under an existing 2020-2030 long-term contract worth US\$500 million would also be suspended. Overall, the downward trend in trade between the two countries, which fell by more than 20 percent to about US\$21 billion in 2022 would likely intensify.¹⁹

CONCLUSION

The statements by South Korea's president have already damaged the country's reputation as a responsible nuclear exporter and, given the current situation on the Korean Peninsula, caused reasonable concern among nuclear nonproliferation experts. However, it is premature to say that Seoul has launched a nuclear weapons program. The recent IAEA's safeguards activities have not detected indications that Seoul is noncompliant with its safeguards agreement.

The current situation once again proves that

19 "South Korea suffers 22.6 percent fall in trade with Russia amid war in Ukraine," *Aju Business Daily*, Feb 22, 2023, www.ajudaily.com/view/20230222105952530

20 "Communication received from the Permanent Representatives of the Russian Federation and the People's Republic of China

concerning a Joint Statement on the Korean Peninsula Issues," International Atomic Energy Agency, Aug. 14, 2017, www.iaea.org/sites/default/files/publications/documents/infircs/2017/infirc922.pdf; "Russia, China coordinate new plan on settlement on Korean Peninsula — diplomat," TASS, Nov. 20, 2019, tass.com/world/1090677

the international community should place the nuclear issue of the Korean Peninsula in a broader context of creating a sustainable regional security architecture, and that the long-term goal of the key stakeholders should be denuclearization of the entire Korean Peninsula. In this regard, there is an urgent need for fresh initiatives that would pave the way to reduce tensions and envisage a possible structure for a regional security architecture. One option here could be to update the plan of action for the comprehensive resolution of the Korean Peninsula issues suggested in the past jointly by Russia and China.²⁰

Overall, one of the first steps toward reducing tensions in the region could be an agreement to suspend joint US and South Korean military exercises in the region in exchange for the suspension of North Korean missile test launches.

Vladislav Chernavskikh is a Research Associate at the Center for Energy and Security Studies (CENESS).

Anton Khlopkov is the Director of the Center for Energy and Security Studies.



"The Morning Peace" published by the Koreapeaceforum will satisfy you

"The Morning Peace" selects and provides columns and press reports which cover unification, South-North relations and Korean peninsula's peace issues in most reasonable and balanced viewpoints in Korea.

If you send your email address to us, we will provide you with regular newsletters.

Subscribe our newsletter by sending email to:
Koreapeaceforum@gmail.com



Introduction to the Korea Peace Forum

"The reunification 'Dream Team' that seeks reconciliation for the two Koreas"

This organization is comprised of former officials from the Ministry of National Unification and diplomats who have planned and executed the engagement policy towards North Korea, scholars who constructed the theoretical foundation for the engagement policy, civil society members who have worked on the ground to lead exchanges between the two Koreas seeking inter-Korean reconciliation and cooperation in all fields.